

1. Lab Overview

In this lab, I created and managed **Organizational Units, Active Directory groups, and delegated permissions** inside a Windows Server 2019 Active Directory environment.

The purpose of this lab was to practice how IT departments organize users, computers, servers, and groups in a domain. I also practiced delegating administrative tasks, which is important for following the **principle of least privilege**.

This lab builds on Lab 04, where I created the [wara.edu](#) domain and joined a Windows 10 client to Active Directory.

2. Business Scenario

This lab simulates a college-style IT environment where different campuses, departments, users, computers, and servers need to be organized inside Active Directory.

As the system administrator, I created a structured OU hierarchy for locations, clients, servers, users, and groups. I also delegated specific administrative responsibilities to built-in groups and created custom groups for different departments.

The original lab section focused on creating OUs, delegating control of OUs, configuring OU permissions, and creating groups.

3. Lab Objectives

The objectives of this lab were to:

- Create top-level Organizational Units.
- Create nested OUs for clients, servers, users, and groups.
- Build a college-style Active Directory structure.
- Move computer objects between OUs.
- Rename and delete OUs.
- Delegate control of OUs.
- Configure granular delegated permissions.
- Configure managed-by information for an OU.
- Create Active Directory groups with different group scopes.

4. Lab Environment

Item	Details
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Server VM	Windows Server 2019
Domain	wara.edu
Main Tool	Active Directory Users and Computers
Main Structure	College-style OU hierarchy
Top-Level OUs	Canandaigua, Geneva, Newark, Victor
Example Groups	IT Level 1, Honors Students, HR Managers
Delegated Groups	Account Operators, Server Operators
Skills Practiced	OU management, group creation, delegation, least privilege

5. Tools Used

- Windows Server 2019
- Active Directory Users and Computers
- Server Manager
- Delegation of Control Wizard
- Active Directory security settings
- Built-in Active Directory groups

6. Steps Performed

Step 1: Created Top-Level Organizational Units

I opened **Active Directory Users and Computers** from Server Manager and created four top-level OUs under the domain:

Canandaigua
Geneva
Newark
Victor

These OUs represented different locations or campuses in the lab environment.

Inside the **Canandaigua** OU, I created additional OUs for:

Administrative
Clients
Servers
Users
Groups

Then I created deeper OU structures under Clients, Servers, Users, and Groups. The lab required creating a detailed OU structure, including Faculty, Staff, Students, server types, and department groups.

Step 2: Created Nested OUs for Clients, Servers, Users, and Groups

Inside the **Clients** OU, I created:

Faculty
Staff
Students

Inside each of those, I created:

Desktops
Laptops

Inside the **Servers** OU, I created server-role OUs such as:

Application Servers
Database Servers
DHCP Servers
DNS Servers
Domain Controllers
File Servers
Print Servers
VPN Servers

Inside the **Groups** OU, I created department-based OUs such as:

Faculty
Finance
Human Resources
Information Technology
Marketing
Staff
Students

This structure helped organize Active Directory objects in a way that would be easier to manage in a real organization.

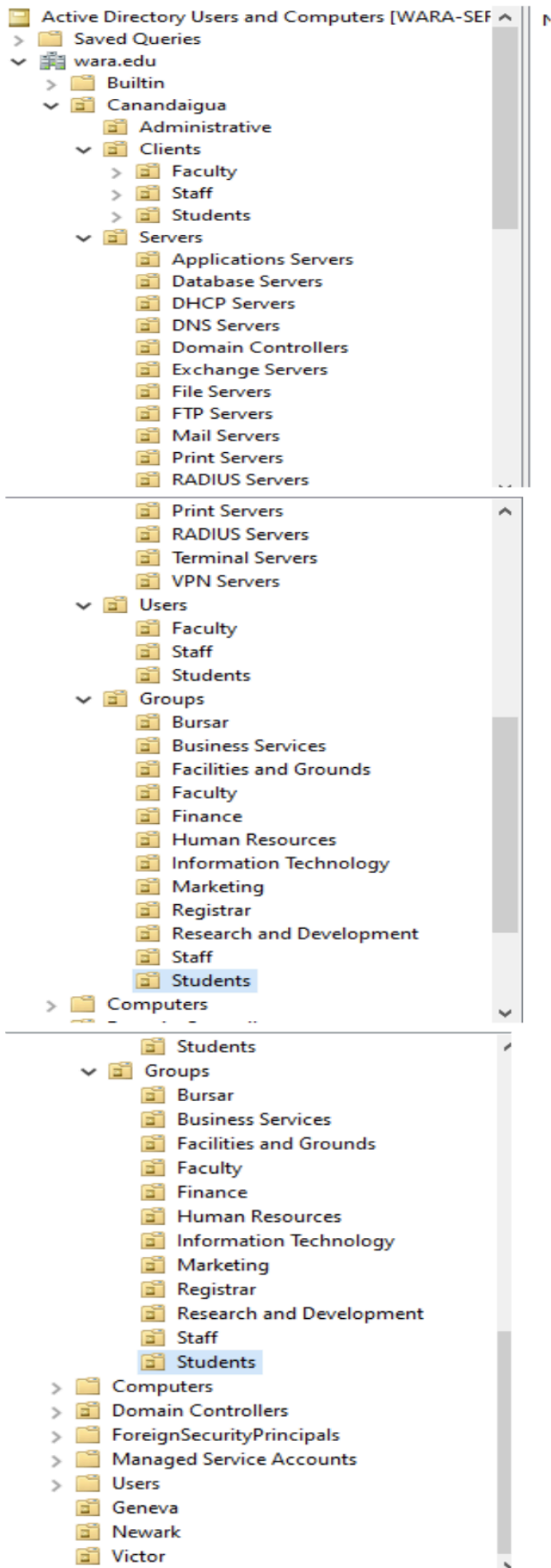


Figure 1: College-style Organizational Unit structure created in Active Directory.

Step 3: Moved, Renamed, and Deleted OUs and Objects

I practiced managing OU objects by:

- Deleting the Bursar OU.
- Renaming the Registrar OU to One Stop Center.
- Moving the client computer object from the default Computers container into the correct OU.
- Moving the computer object between Desktops and Laptops.
- Using drag-and-drop and cut-and-paste methods to move objects.

This helped me understand how computer objects can be organized after they are joined to a domain.

Active Directory Users and Computers [WARA-SEF ^		Name	Type
- Saved Queries - wara.edu - Built-in - Canandaigua - Administrative - Clients - Faculty - Desktops - Laptops - Staff - Desktops - Laptops - Students - Desktops - Laptops - Servers - Applications Servers - Database Servers - DHCP Servers - DNS Servers - Domain Controllers - Exchange Servers - File Servers - FTP Servers - Mail Servers - Print Servers - RADIUS Servers - Terminal Servers - VPN Servers - Users - Faculty - Staff - Students		WARA-CLIENT	Computer

		Name	Type
- Domain Controllers - Exchange Servers - File Servers - FTP Servers - Mail Servers - Print Servers - RADIUS Servers - Terminal Servers - VPN Servers - Users - Faculty - Staff - Students - Groups - Business Services - Facilities and Grounds - Faculty - Finance - Human Resources - Information Technology - Marketing - One Stop Center - Research and Development - Staff - Students - Computers - Domain Controllers - ForeignSecurityPrincipals - Managed Service Accounts - Users - Geneva - Newark - Victor		WARA-CLIENT	Computer

Figure 2: Computer object moved into the proper Organizational Unit.

Step 4: Delegated Control of the Users OU

Next, I used the **Delegation of Control Wizard** to delegate password-related tasks for the **Users** OU.

I selected the **Account Operators** group and delegated these common tasks:

Reset user passwords

Force password change at next login

This allowed delegated users to perform specific account support tasks without giving them full domain administrator access.

The lab also explained that using default groups like Account Operators may provide more permissions than needed. A better real-world practice is to create a custom group with only the exact permissions required.

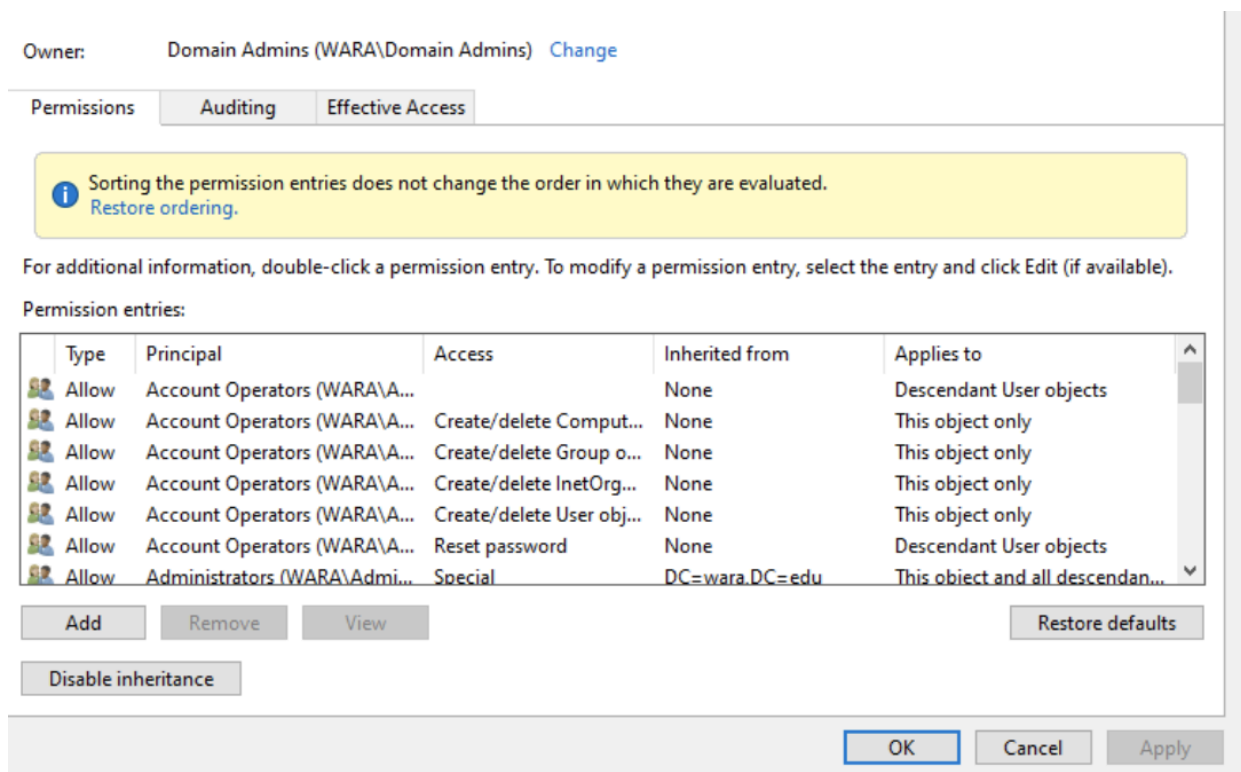


Figure 3: Delegated password reset permissions for the Users OU.

Step 5: Performed Granular Delegation for the Servers OU

I also practiced more specific delegation on the **Servers** OU.

I selected the **Server Operators** group and created a custom delegated task for **Computer Objects**. The delegated permissions included the ability to:

- Create selected computer objects.
- Delete selected computer objects.
- Read permissions.
- Write permissions.

This showed how administrators can delegate only certain tasks to specific groups instead of giving broad administrative access.

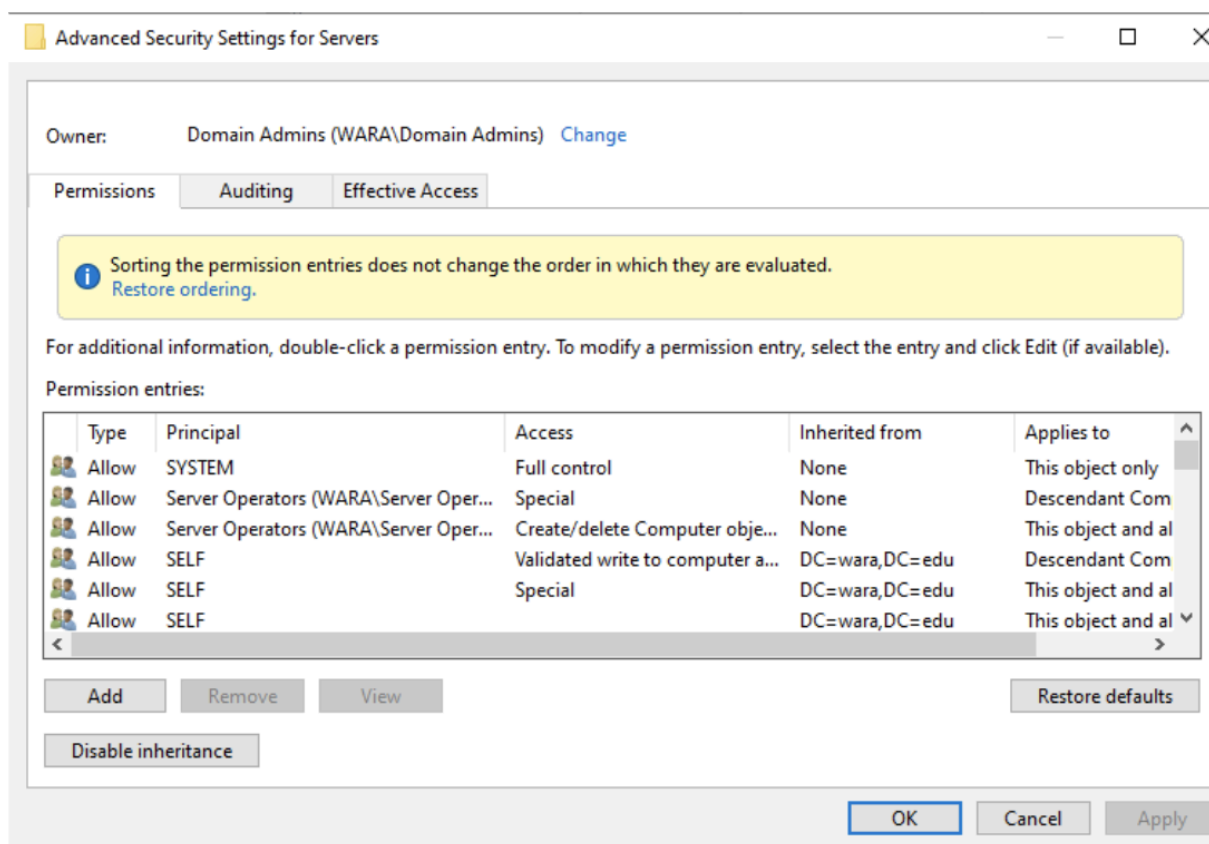


Figure 4: Custom delegation configured for computer objects in the Servers OU.

Step 6: Configured Managed By Information for an OU

I configured contact information for the **Victor** OU by using the **Managed By** tab and assigning the Administrator account.

This is useful in real environments because it helps identify who is responsible for managing or owning a specific OU or department.

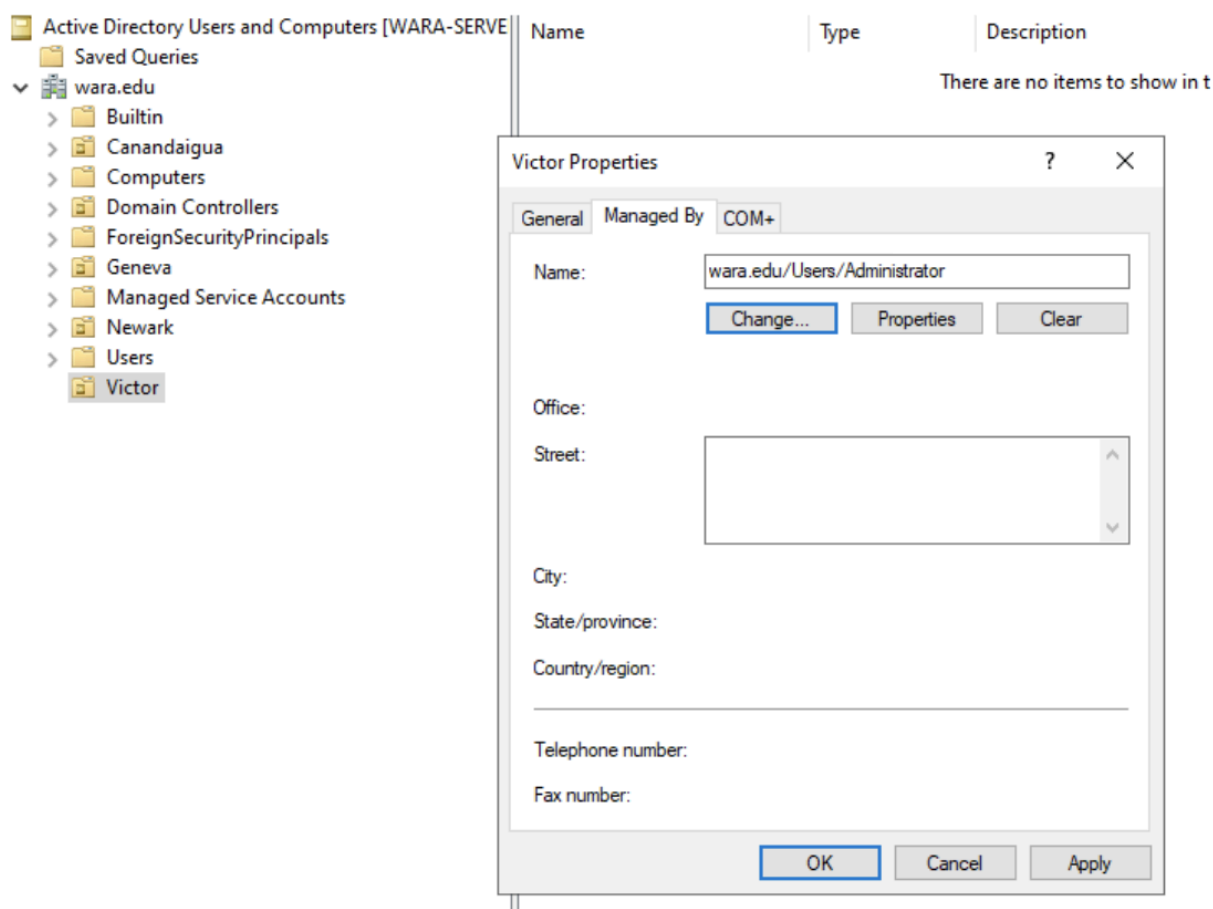


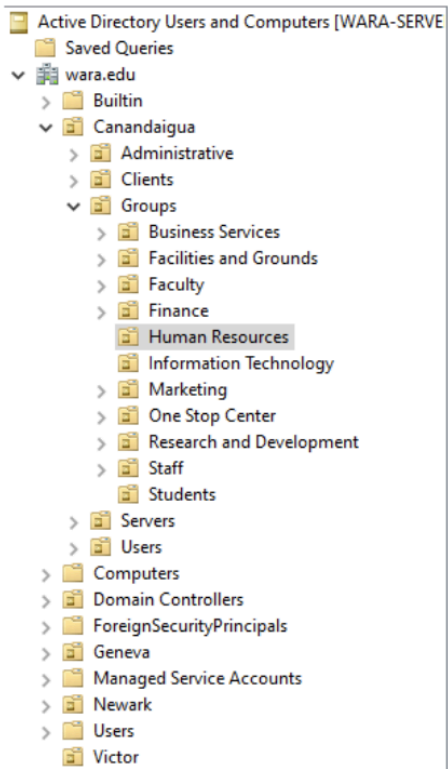
Figure 5: Managed By information configured for the Victor OU.

Step 7: Created Active Directory Groups

Finally, I created several groups inside the correct department OUs:

Group Name	Location	Group Scope
IT Level 1	Information Technology OU	Global
Honors Students	Students OU	Domain Local
HR Managers	Human Resources OU	Universal

The lab included creating these groups with different group scopes, which helped me understand how groups are used to organize access and responsibilities inside Active Directory.



Name	Type	Description
HR Managers	Security Group - Global	

HR Managers Properties

General Members Member Of Managed By

 HR Managers

Group name (pre-Windows 2000):

Description:

Email:

Group scope

☐ Domain local

☒ Global

☐ Universal

Group type

☒ Security

☐ Distribution

Notes:

OK Cancel Apply

- Active Directory Users and Computers [WARA-SERVE]
 - Saved Queries
 - wara.edu
 - Builtin
 - Canandaigua
 - Administrative
 - Clients
 - Groups
 - Business Services
 - Facilities and Grounds
 - Faculty
 - Finance
 - Human Resources
 - Information Technology
 - Marketing
 - One Stop Center
 - Research and Development
 - Staff
 - Students
 - Servers
 - Users
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Geneva
 - Managed Service Accounts
 - Newark
 - Users
 - Victor

Name	Type	Description
Honors Students	Security Group - Global	

Honors Students Properties

General Members Member Of Managed By

 Honors Students

Group name (pre-Windows 2000):

Description:

E-mail:

Group scope

☐ Domain local

☒ Global

☐ Universal

Group type

☒ Security

☐ Distribution

Notes:

OK Cancel Apply

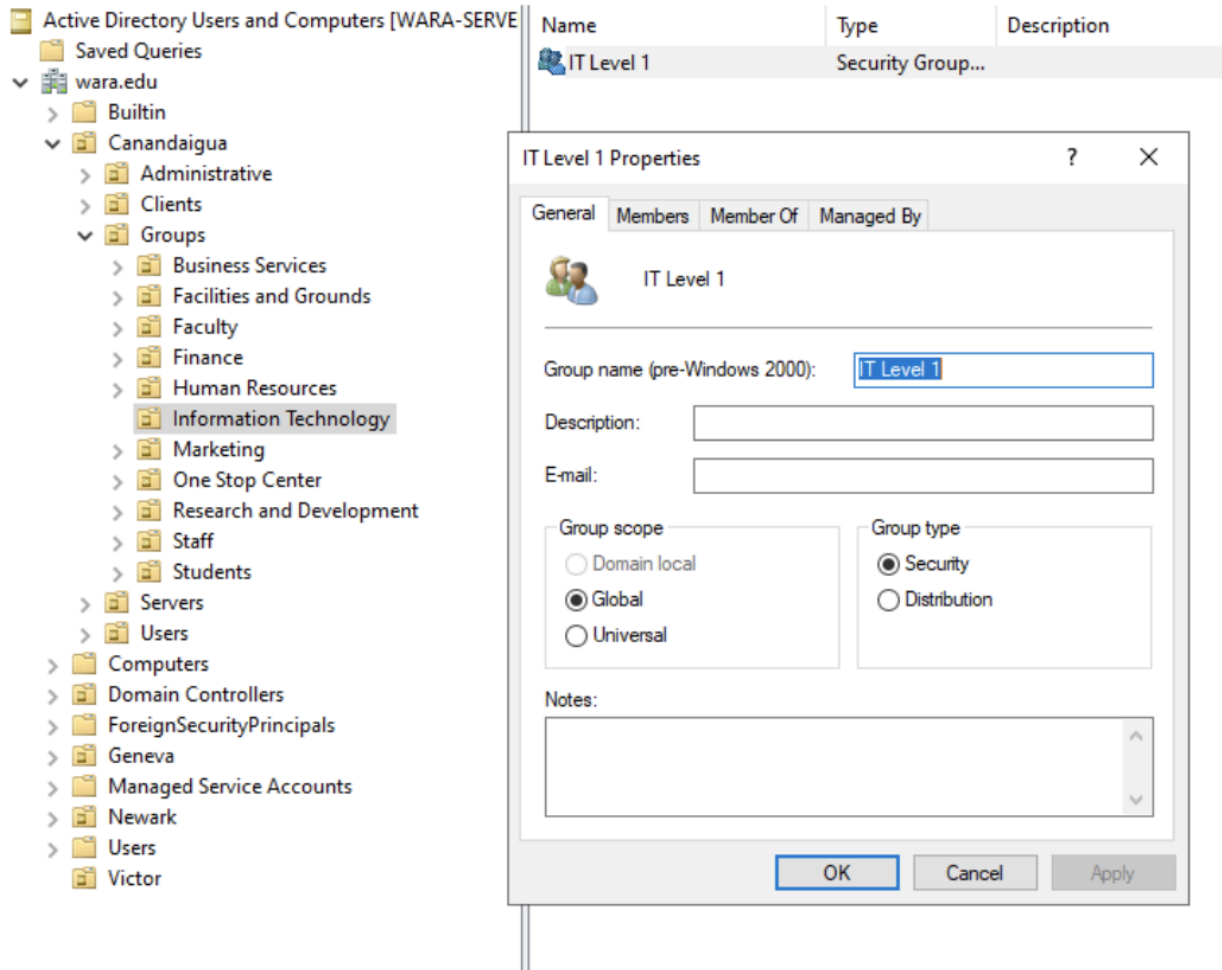


Figure 6: Active Directory security group created inside the correct department OU.

7. Key Findings

Finding	Explanation
OUs help organize Active Directory	OUs make it easier to manage users, computers, servers, and groups.
Nested OUs support real business structure	Departments, campuses, and device types can be separated logically.
Delegation reduces admin workload	Specific tasks can be assigned to support teams without full admin access.
Least privilege matters	Custom groups are better than broad built-in groups when possible.

Group scopes are important	Global, Domain Local, and Universal groups serve different purposes.
Object movement needs careful planning	Moving users or computers affects organization and policy application.

8. Analysis

This lab showed how Active Directory can be organized to support a real business or school environment. Creating OUs helps administrators keep objects organized and apply policies more easily.

Delegation was one of the most important parts of this lab. Instead of giving every IT staff member full domain administrator access, specific tasks can be delegated. For example, help desk staff may only need permission to reset passwords or force password changes at next logon.

This is important because excessive permissions can create security risks. A well-designed Active Directory structure should support both efficient administration and least privilege security.

9. IT Support Relevance

This lab is directly useful for IT Support Analyst, Help Desk, Desktop Support, and Junior System Administrator roles.

It connects to real job tasks such as:

- Finding users and computers in Active Directory
- Understanding OU structure
- Moving computer objects to the correct OU
- Creating and managing groups
- Understanding delegated permissions
- Supporting password reset workflows
- Following least privilege principles
- Escalating permission issues properly
- Documenting Active Directory changes

10. Skills Demonstrated

This lab demonstrates:

- Active Directory Users and Computers
- Organizational Unit creation
- Nested OU structure design
- Computer object management
- Group creation
- Group scope understanding
- Delegation of Control Wizard
- Custom delegation
- Least privilege access
- Active Directory administration
- Technical documentation

11. Ticket-Style Summary

Issue:

The domain environment needed an organized Active Directory structure for users, computers, servers, groups, and delegated administration.

Impact:

Without a proper OU and group structure, managing users, computers, policies, and permissions would be inefficient and less secure.

Steps Taken:

- Created top-level OUs for different locations.
- Built nested OUs for clients, servers, users, and groups.
- Created department-based OUs.
- Moved computer objects into the correct OUs.
- Renamed and deleted test OUs.
- Delegated password reset permissions for the Users OU.
- Configured custom delegation for computer objects in the Servers OU.
- Added managed-by contact information for an OU.
- Created IT Level 1, Honors Students, and HR Managers groups.

Root Cause:

The Active Directory environment needed a logical structure and delegated permissions for easier administration.

Resolution:

Created a structured OU hierarchy, configured delegation, and created security groups for administrative and department-based organization.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learned that Active Directory organization is very important in a real IT environment. OUs are not just folders; they help organize users, computers, servers, and groups so administrators can manage them more efficiently.

I also learned that delegation is important because it allows specific support tasks to be assigned without giving full administrator privileges. This supports better security and follows the principle of least privilege.